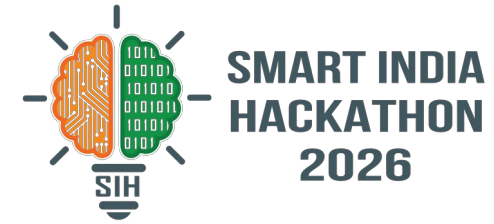


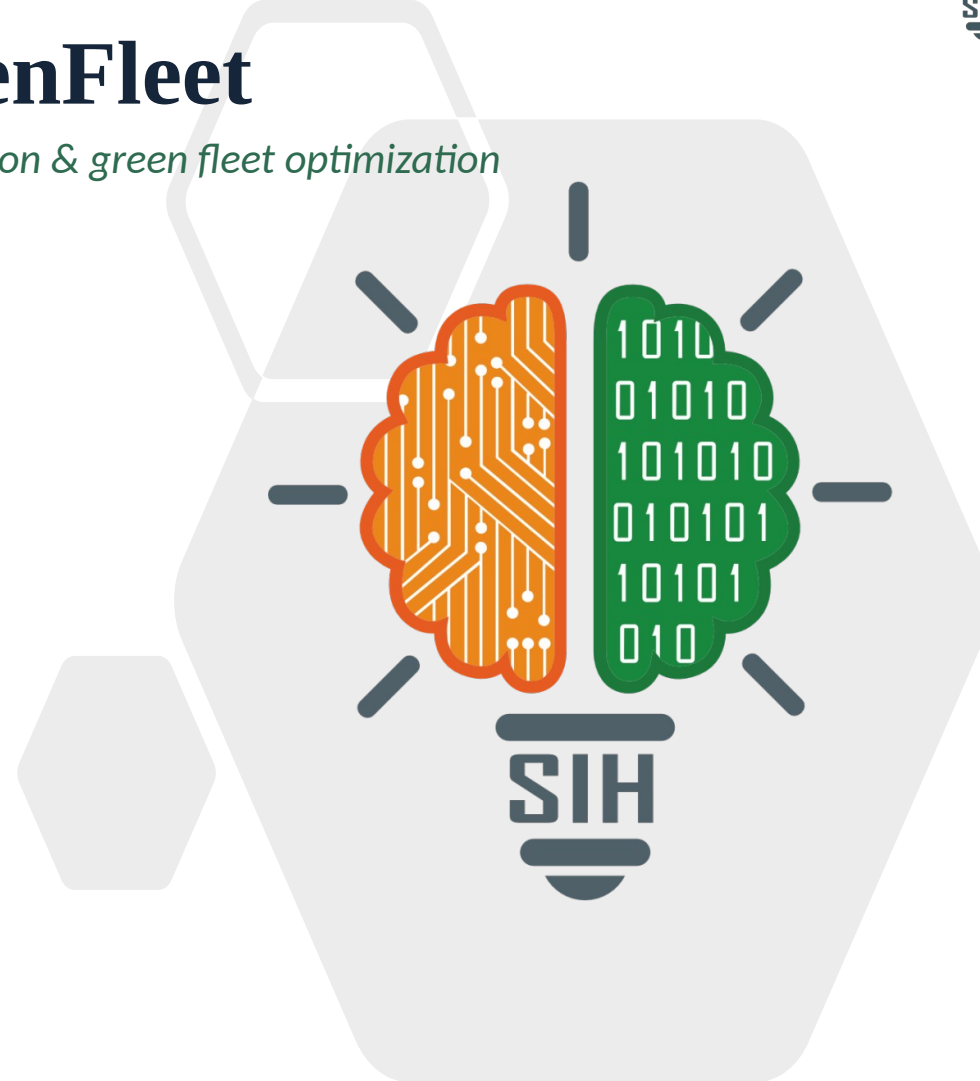
SMART INDIA HACKATHON 2026



Q-GreenFleet

Quantum-inspired fuel prediction & green fleet optimization

- **Problem Statement ID** – 26138
- **Problem Statement Title** – Quantum-Inspired Fuel Consumption Prediction and Green Fleet Optimization
- **Theme** – Clean & Green Technology
- **PS Category** – Software
- **Team ID** –
- **Team Name (Registered on portal)** –

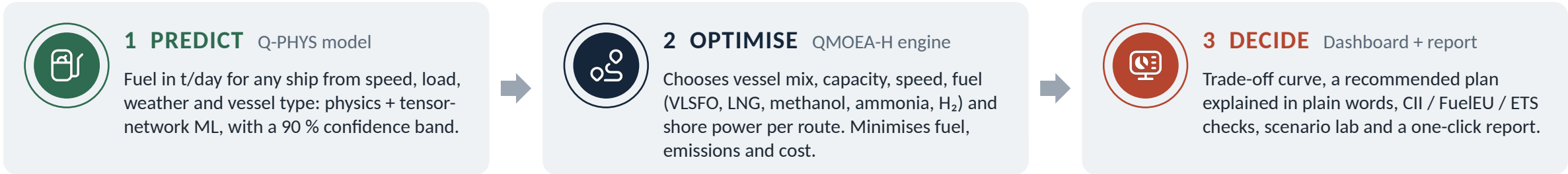




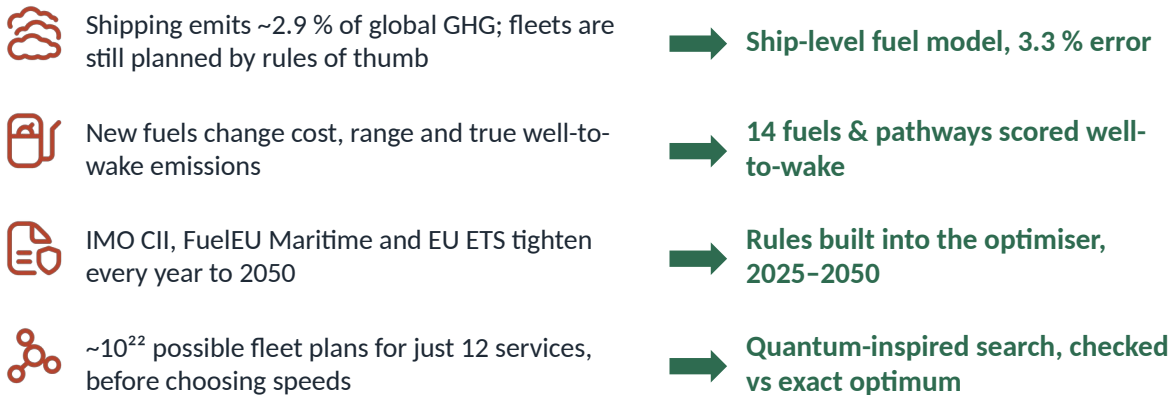
Proposed Solution (Describe your Idea/Solution/Prototype)

One decision engine that **predicts every ship's fuel burn** and **re-plans the whole fleet** (vessel mix, capacity, speed, fuel and shore power) for the lowest fuel, emissions and cost that still meet cargo demand, schedules and emission rules.

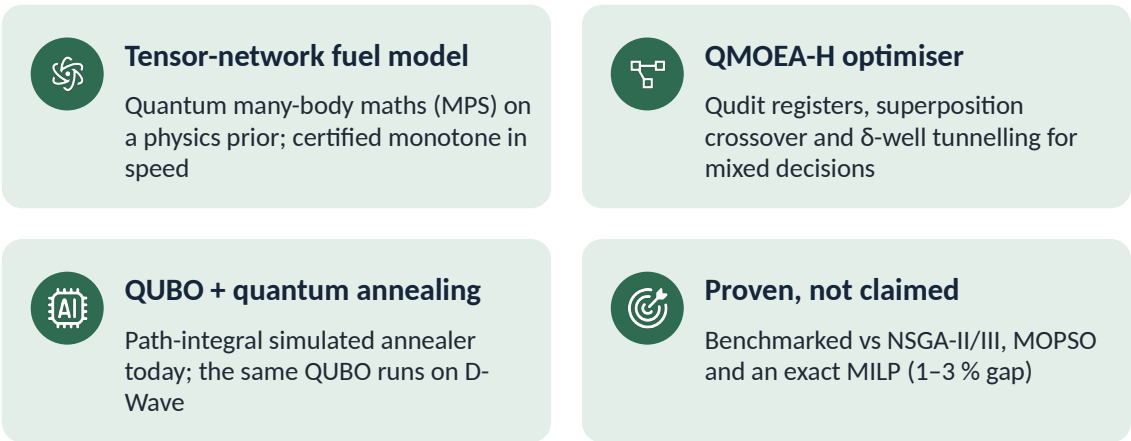
DETAILED EXPLANATION OF THE PROPOSED SOLUTION



HOW IT ADDRESSES THE PROBLEM



INNOVATION AND UNIQUENESS OF THE SOLUTION



TECHNICAL APPROACH

Technologies to be used

languages · frameworks · hardware



Core & data

Python 3.11 · NumPy · pandas · searoute



Prediction (ML)

LightGBM · scikit-learn · own MPS tensor network



Quantum-inspired & exact

pymoo · PuLP/CBC MILP · OpenJij · dimod (D-Wave)



Backend

FastAPI · Uvicorn · live jobs over SSE



Frontend

React · TypeScript · Vite · Plotly · Leaflet



DevOps

Docker · GitHub Actions CI · pytest · Playwright

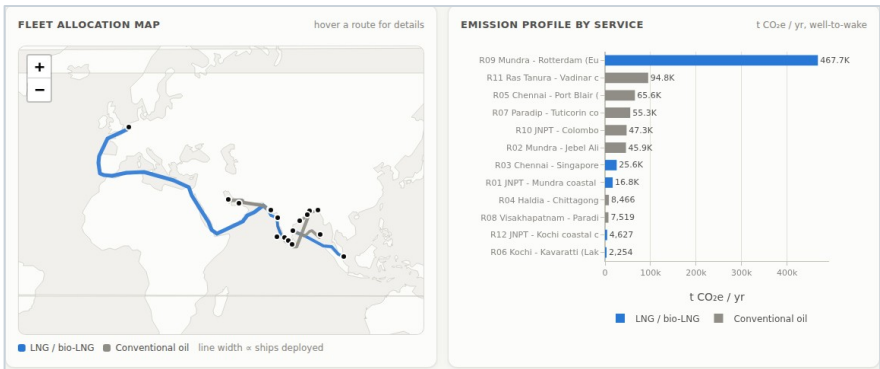


Hardware

Any laptop CPU. No GPU, no qubits needed

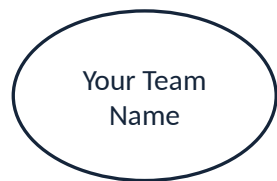
Methodology and process for implementation

flow chart · working prototype

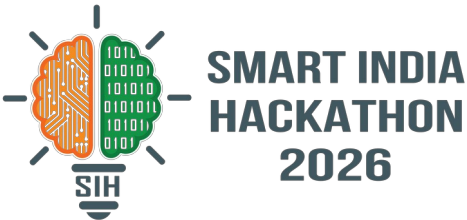


WORKING PROTOTYPE

- ✓ Built and running today (India & EU networks)
- ✓ Live Pareto front, fleet map, emission profiles
- ✓ 71 automated tests · CI green
- ✓ ~55,000 fleet plans evaluated per second
- ✓ One command: docker compose up



FEASIBILITY AND VIABILITY



ANALYSIS OF THE FEASIBILITY OF THE IDEA

3.3 %

fuel prediction error (MAPE) on held-out voyages

55k/s

fleet plans evaluated per second on an ordinary CPU

1.4 s

to solve the India network exactly (MILP), our accuracy yardstick

71

automated tests, CI green, one-command Docker run

0

GPUs or qubits needed today; D-Wave-ready later

⚠️ POTENTIAL CHALLENGES AND RISKS

Little real data for LNG, ammonia and H₂ ships

ML can output unphysical fuel curves that an optimiser exploits

Fuel prices and regulations keep shifting

Quantum hardware is still immature

Very large fleets (100+ routes) converge more slowly

✅ STRATEGIES FOR OVERCOMING THESE CHALLENGES

Physics prior + fleet head: 6.0 % error even on unseen vessel types; loaders for FuelCast, EU MRV and operator logs

Booster certified non-decreasing in speed; 90 % conformal intervals flag uncertainty

Scenario lab (2025–2050, prices, carbon, Red Sea, monsoon) + Monte Carlo robustness; factors in editable YAML

Quantum-inspired algorithms run on normal CPUs now; the same QUBO runs on D-Wave later

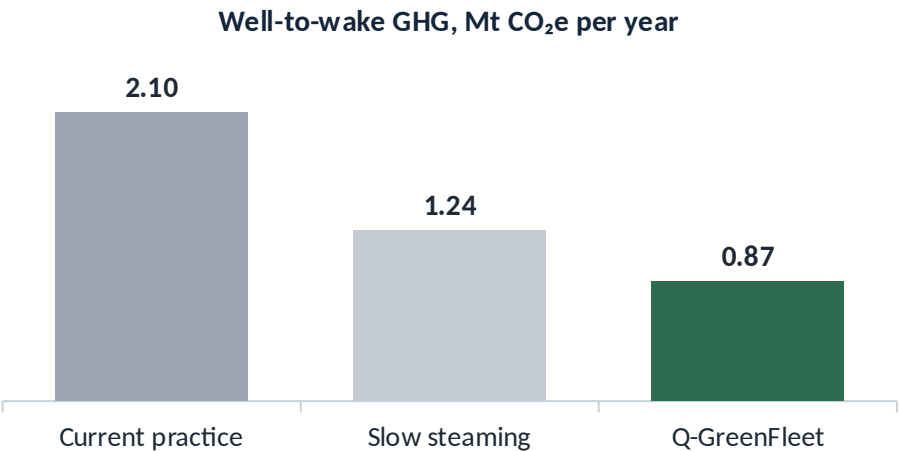
Every instance certified by the exact MILP; warm starts and adaptive rates on the roadmap



Viability: SaaS or on-premise licence for fleet operators and ports. On the India network one avoided FuelEU penalty (\$21.7 M a year) plus a 25 % lower fleet cost far outweighs running costs; the stack is open source and CPU-only.

IMPACT AND BENEFITS

INDIA 2030 CASE STUDY • 12 SERVICES



-59 % GHG

1.23 Mt CO₂e a year avoided

-52 % fuel

297 kt a year less fuel (HFO-equivalent)

-25 % cost

\$179 M a year lower fleet cost

12/12 CII A-B

services compliant (1 of 12 today);
\$0 FuelEU penalty

POTENTIAL IMPACT ON THE TARGET AUDIENCE



Shipping lines & fleet operators

SCI, coastal and near-sea lines: lower fuel bills, CII / FuelEU compliance, ready reports



Ports: Sagarmala & Harit Sagar green ports

plan shore power and green-fuel bunkering demand



Regulators: DG Shipping, MoPSW

evidence for policy; test 2025–2050 pathways before they bind



Green H₂ / ammonia hubs: Kandla, Tuticorin, Paradip

see when and where green-fuel demand arrives

BENEFITS OF THE SOLUTION



Environmental

- 1.23 Mt CO₂e a year less on 12 services
- Shore power on 10 services: cleaner port air
- Clear pathway to LNG and e-ammonia by 2050



Economic

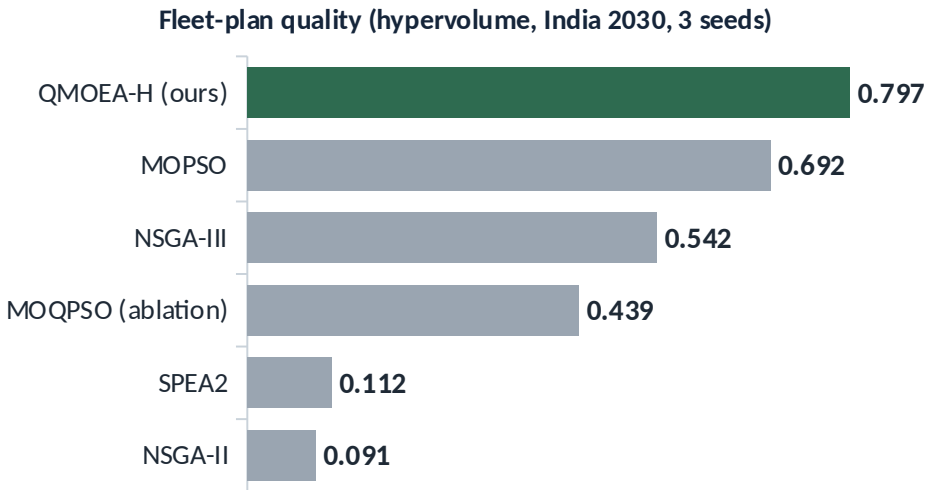
- \$179 M a year lower fleet cost
- No FuelEU penalty; smaller EU ETS bill
- Less stranded-asset risk as rules tighten



Social

- Backs Maritime India Vision 2030 and net zero 2070
- Skills and jobs in green-fuel shipping
- Healthier air in port cities

OUR RESEARCH & EXPERIMENTS



 **Prediction:** 3.29 % MAPE vs 5.28 % LightGBM; Wilcoxon $p \leq 0.001$ on 20 ships

 **Exact gap:** 1.1 % fuel · 3.1 % GHG · 3.2 % cost from the MILP optimum

 **QUBO + SQA:** our path-integral annealer best in 4 of 6 cases; every plan feasible

 **Honest limit:** MOPSO leads on loose 50–100-route networks: on our roadmap

Details / Links of the reference and research work

MARITIME PHYSICS & REGULATION

1. IMO (2020). Fourth IMO GHG Study: bottom-up ship power and fuel model
2. Regulation (EU) 2023/1805, FuelEU Maritime, Annex II well-to-wake factors
3. IMO MEPC.353(78), MEPC.338(76), MEPC.400(83): CII reference lines & reduction factors
4. Directive (EU) 2023/959: EU ETS extended to shipping
5. Kwon, Y. J. (2008). Speed loss due to added resistance in wind and waves. The Naval Architect

ALGORITHMS

6. Stoudenmire & Schwab (2016). Supervised learning with tensor networks. NeurIPS
7. Sun, Feng & Xu (2004). PSO with particles having quantum behavior. IEEE CEC
8. Han & Kim (2002). Quantum-inspired evolutionary algorithm. IEEE Trans. Evol. Comput.
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10. Lucas (2014). Ising formulations of many NP problems. Frontiers in Physics
11. Deb et al. (2002). NSGA-II. IEEE Trans. Evol. Comput.
12. Angelopoulos & Bates (2021). A gentle introduction to conformal prediction. arXiv:2107.07511

DATA & INDIAN POLICY

13. FuelCast dataset (huggingface.co/datasets/krohn.digital/FuelCast); EU MRV / THETIS-MRV (EMSA)
14. MoPSW (2023). Harit Sagar Green Port Guidelines; Maritime India Vision 2030



Prototype code & benchmark reports: github.com/DeepakSinghhh/PS-138

Icons: Ulcons by Flaticon (flaticon.com) · Line Awesome by Icons8 (icons8.com)